

Prachi Patolia

703-332-1901 | patolia.prachi26@gmail.com | [LinkedIn](#)

EDUCATION

Georgetown University, Washington DC - MS, Health Informatics and Data Science	Aug 2024
California State University, San Bernardino - PG Diploma, Hospital Administration	Jan 2022
Darshan Dental College and Hospital, Rajasthan, India - Bachelor, Dental Surgery	Oct 2019

WORK EXPERIENCE

Freelance, Remote - Data Scientist	Sep 2024 — Feb 2025
Predictive Modeling: Built and optimized machine learning models (Random Forest, Logistic Regression, XGBoost) using CDC obesity risk data, achieving 97% predictive accuracy. Data Engineering Efficiency: Automated preprocessing workflows with SimpleImputer and OneHotEncoder, reducing data cleaning time by 80% and accelerating model deployment. Impactful Public Health Insights: Delivered actionable findings to a non-profit, driving a targeted awareness campaign that reached over 10K high-risk individuals across multiple regions. Interactive Data Visualization: Created dynamic Tableau dashboards highlighting obesity hotspots, enabling real-time decision-making and localized intervention strategies for stakeholders.	
ICF, Virginia - Data Scientist Intern	May 2024 — Aug 2024
AI Chatbot Development: Engineered a generative AI chatbot for PDC using GraphQL APIs, Weaviate, and LLMs (BioMistral, Llama 3), enhancing response accuracy via Retrieval Augmented Generation (RAG). LLM-Based Embedding: Leveraged Hugging Face's PubMedBERT to preprocess and embed complex scientific/medical text, significantly reducing manual search time by 70% for researchers and medical staff. Vector Database Integration: Integrated Weaviate for real-time embedding storage and retrieval, enabling faster access to insights and reducing response latency by 45% across the system. Biomedical NLP Optimization: Fine-tuned domain-specific LLMs on curated clinical vocabularies to generate context-aware replies, improving clinical content relevancy and engagement by 60%. Cross-Functional Collaboration: Partnered with cloud, NLP, and public health teams to align chatbot functionality with end-user goals, leading to 30% higher engagement in pilot rollout.	
Asian Bariatrics Hospital, Ahmedabad, India - Data Analyst	Nov 2020 — Jul 2023
Operational Efficiency Analytics: Used advanced SQL to analyze EMR and administrative data, tracking KPIs that led to a 35% reduction in patient wait times and a 22% improvement in doctor utilization rates. Data Visualization & Insights: Designed actionable Tableau dashboards visualizing patient satisfaction, treatment trends, and revenue streams, used weekly by 10+ department heads. Process Optimization: Led cross-functional Lean Six Sigma initiatives, including SPC charting and root cause analysis, improving admission throughput and pushing patient satisfaction to 95%. NABH Accreditation Leadership: Orchestrated end-to-end NABH accreditation process, contributing to a 25% surge in monthly patient intake and >INR 3.5M revenue increase over 12 months. Team Management: Supervised scheduling and clinical performance of 15+ medical officers, improving documentation turnaround time by 40% and ensuring regulatory compliance.	
Marengo CIMS Hospital, Ahmedabad, India - Associate Dentist	Nov 2019 — Oct 2020
Patient Care: Delivered supervised, patient-focused dental treatments, including exams, cleanings, fillings, and extractions, while ensuring safety and clear patient communication. Clinical Documentation: Maintained accurate clinical records to support treatment planning, regulatory compliance, and effective communication among healthcare teams. Team Collaboration: Worked closely with senior dentists and multidisciplinary support staff to improve clinic workflows, significantly boosting overall efficiency and patient satisfaction.	

PROJECTS

EMR Workflow Enhancement for Hypertension Management
Redesigned EMR usability for hypertension care using synthetic Synthea data by implementing clinical forms and critical threshold alerts. Developed Tableau dashboards to track patient metrics in real time, resulting in a 20% reduction in time-to-treatment initiation.
Maternity Application - Patient & Provider Facing
Created a maternity care app prototype using Figma, featuring appointment scheduling, EHR access, and wearable device integration. Applied feedback from 50+ users to improve usability by 35% and enhance patient engagement.

SKILLS

Programming & Scripting	: Python, R, SQL, JavaScript
Data Science & Machine Learning	: Scikit-learn, XGBoost, Random Forest, Logistic Regression, GridSearchCV, Pandas, NumPy, enrichR, SimpleImputer, OneHotEncoder, 5-fold Cross-Validation
Data Visualization & BI Tools	: Tableau, Microsoft Power BI, Excel (PivotTables, VLOOKUP), Matplotlib, Seaborn
Healthcare Technologies	: HL7, FHIR, OMOP CDM, RedCap, HL7 Soup, Synthea
NLP & AI/LLMs	: Hugging Face Transformers, PubMedBERT, BioMistral, Llama 3, Retrieval-Augmented Generation (RAG), Weaviate, Vector Embeddings
Tools & Platforms	: Jupyter Notebook, RStudio, Git, Docker, SevenBridges, Pentaho
Cloud & Big Data	: Google Cloud Platform (GCP), Microsoft Azure, Hadoop
Project & Collaboration Tools	: Figma, Jira, Confluence